🎓 AlumniConnect

Intelligent Platform to Bridge Alumni & Students

📋 Project Q&A — Interview / Presentation Sheet

1. PROJECT OVERVIEW

1 What is your project?

AlumniConnect is a full-stack web platform that connects college alumni and students through an intelligent bridge. It includes features like AI-powered mentorship matching, a career guidance chatbot, interview practice, a job board, and a discussion forum.

2 Why did you build this project?

Students often struggle to find senior alumni for career guidance. There is no dedicated platform for job referrals, mock interviews, and skill roadmaps. AlumniConnect solves all these problems in one place with AI assistance, available 24/7.

3 Who will use this project?

- **Students** — For career guidance, interview practice, job search, and finding mentors
- **Alumni** — For posting jobs, providing mentorship, and contributing to the community
- **Admin** — For managing the platform and verifying users
- **Colleges / Universities** — For digitizing their alumni network

4 What main problem does this project solve?

- Students lack proper career guidance — the AI chatbot is available 24/7
- Alumni networks remain disconnected — the platform keeps everyone connected
- No dedicated mock interview platform — AI-generated questions and evaluation are available
- Students can't connect for job referrals — an integrated job board is available

2. TECHNOLOGY STACK

5 What technologies have you used? (Tech Stack)

Layer	Technology
Frontend	React 18, TypeScript, Vite, Tailwind CSS
Backend	Node.js, Express.js, TypeScript
Authentication	Firebase Authentication (Google + Email/Password)
Database	Firebase Firestore + Firebase Realtime Database
AI Model	Groq API — LLaMA 3.3 70B Versatile
Frontend Hosting	Vercel
Backend Hosting	Render

6 Which AI model have you used and why?

We used the **Groq API** which runs the **LLaMA 3.3 70B Versatile** model.

- **LLaMA 3.3 70B** — Meta's open-source, extremely powerful language model with 70 billion parameters
- **Groq** — The world's fastest AI inference platform; responses come in milliseconds
- **Free Tier** — Completely free for students with high rate limits
- Career guidance, interview evaluation, skill roadmap — all powered by this model

Model name: `llama-3.3-70b-versatile`

7 Why use Firebase and not any other database?

- Firebase's free tier includes a real-time database, authentication, and cloud storage
- Google Sign-In integrates in a single click
- Firebase Realtime Database is perfect for real-time chat — without needing a WebSocket server
- Setup is very fast — no server management needed

8 Why choose React for the frontend?

React 18 is component-based and allows building reusable UI pieces. TypeScript provides type safety, reducing bugs. Vite makes the build process very fast. Tailwind CSS enables quick responsive UI development without writing custom CSS.

3. FEATURES — DEEP DIVE

9 What features does the project have?

🔐

Authentication

Email/Password + Google Sign-in, Role-based access

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AI Mentor Matching

AI suggests the best mentor based on the student's profile

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Career Chatbot

24/7 AI career guidance, multi-turn conversation

🎯

Interview Practice

AI generates questions and evaluates answers with a score

💼

Job Board

Alumni post jobs/internships, students apply

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Discussion Forum

Community Q&A, upvotes, tags

💬

Real-time Chat

1-on-1 messaging between student & alumni

🗺️

Skill Roadmap

AI generates a personalized learning path for any skill

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Resume Review

AI ATS scoring + improvement suggestions

🔍

Skill Gap Analysis

AI tells you what to learn based on your current skills

📊

Dashboard

Analytics — user growth, charts, statistics

🛡️

Admin Panel

User management, content moderation

10 How does the Career Chatbot work?

User sends a message → Backend verifies the Firebase token → Message is sent to the Groq API (LLaMA 3.3 70B) with conversation history → AI response is received → Displayed on the frontend. The complete conversation history is maintained for multi-turn chat.

11 How does the Interview Practice feature work?

- Student chooses a **topic** (e.g., React, Python) and **difficulty** (easy/medium/hard)
- AI **generates 5 questions** in JSON format
- Student types the answer for each question
- AI **evaluates each answer** — providing a score (0–10), strengths, weaknesses, and a model answer
- An overall performance summary is given at the end

12 What is the Skill Roadmap feature?

The student types any skill (e.g., "Machine Learning", "Web Development") and AI generates a **step-by-step personalized learning roadmap** — with resources, timeline, and priority. The roadmap is also saved on the dashboard.

4. ARCHITECTURE & SECURITY

13 What is the project architecture?

3-tier architecture:

Tier 1 — Frontend (Vercel): React app — UI, routing, state management
Tier 2 — Backend (Render): Express API — business logic, AI calls, security middleware
Tier 3 — Database (Firebase): Firestore (structured data) + Realtime DB (chat)

Frontend → Backend REST API → Firebase DB + Groq AI

14 What have you done for security?

- **Firebase Authentication** — JWT token is verified with every request
- **Helmet.js** — HTTP security headers are automatically set
- **Rate Limiting** — 500 requests per 15 min per user; 50 req/min for AI
- **CORS** — only the allowed frontend URL can access the backend
- **Environment Variables** — no secret keys are hardcoded in the code
- **Firestore Security Rules** — users can only read/write their own data

15 How does authentication work?

User logs in (Google or Email/Password) → Firebase Authentication provides a JWT token → Frontend sends this token in the Authorization header with every API call → Backend `verifyToken` middleware verifies the token using Firebase Admin SDK → only then is the data returned.

16 How is role-based access implemented?

During signup, the user chooses their role: **Student / Alumni / Admin**. This role is stored in the user document in Firebase Firestore. The Admin panel is visible only to users with the Admin role. Alumni can post jobs; students cannot.

5. DEPLOYMENT & SETUP

17 Where is the project deployed?

- **Frontend** — Deployed on **Vercel** (free tier, automatic deployment from GitHub)
- **Backend** — Deployed on **Render** (free tier Node.js server)
- **Database** — **Firebase** (on Google Cloud, free tier)
- Automatically re-deployed whenever code is pushed to GitHub

18 What are the steps to run the project locally?

Prerequisites: Node.js 18+, Firebase project, Groq API key

Backend:
`cd backend` → `npm install` → add keys to `.env` file → `npm run dev`

Frontend:
`cd frontend` → `npm install` → add Firebase config to `.env` file → `npm run dev`

In browser: `http://localhost:3000`

19 Can the project be run completely for free?

Yes, 100% free!

- Groq API — very high limits in the free tier
- Firebase — free Spark plan is sufficient for students
- Vercel — unlimited deployments in the free tier
- Render — backend hosting in the free tier

Cost: $0 / month for normal usage.

6. AI / TECHNICAL DEEP DIVE

20 What is the difference between Groq API and Gemini API? Why choose Groq?

Feature	Groq API	Google Gemini
Speed	⚡ Extremely fast (LPU chips)	Moderate
Model	LLaMA 3.3 70B (Meta)	Gemini Flash/Pro
Free Tier	Very generous limits	Limited
Open Source	Yes (LLaMA)	No (closed)
API Format	OpenAI-compatible	Custom format

Groq was chosen because it is the fastest and provides high limits in the free tier.

21 What is the LLaMA 3.3 70B model?

LLaMA (Large Language Model Meta AI) is Meta's open-source AI model. **70B** means 70 billion parameters — it is comparable to ChatGPT in intelligence. The "Versatile" variant is optimized for general-purpose tasks — coding, writing, analysis, and question answering.

22 How is real-time chat implemented?

Firebase Realtime Database is used. When a user sends a message, it is saved in Firebase at the `/chats/{chatId}/messages` path. Firebase's `onValue` listener on both users' frontends listens for real-time updates — as soon as a message arrives, it is instantly shown on screen without a page refresh.

23 Why use TypeScript instead of JavaScript?

TypeScript provides **type safety** — runtime errors are caught at compile time. Bugs are reduced in large codebases, better autocomplete is available in the IDE, and team collaboration becomes easier.

7. CHALLENGES & LEARNINGS

24 What challenges did you face while building the project?

- **CORS issues** — Frontend and backend are on different domains; CORS had to be configured
- **Rate limiting** — AI API calls need to be throttled in the free tier
- **Firebase private key** — There was a newline encoding issue in environment variables
- **Real-time chat sync** — Ensuring message ordering between multiple users
- **AI response parsing** — Consistently extracting interview questions in JSON format

25 How can this project be improved further? (Future Scope)

- 📱 **Mobile App** — Android/iOS app using React Native
- 🎥 **Video Mentorship** — Video calling using WebRTC
- 📊 **Analytics Dashboard** — Alumni engagement metrics
- 🌐 **Multi-language support** — Hindi, regional languages
- 📄 **Resume Builder** — AI-powered resume generation
- 🏆 **Gamification** — Points, badges for active users
- 📧 **Email Notifications** — New messages, job alerts

26 What did you learn from this project?

- Full-stack development — React, Node.js, and TypeScript together
- Firebase Authentication + Firestore + Realtime DB integration
- Production-grade AI API integration with error handling and rate limiting
- Deployment pipeline — Vercel (frontend) + Render (backend)
- Security best practices — JWT, CORS, Helmet, rate limiting
- Real-time features without WebSocket (Firebase listener)

8. QUICK REFERENCE

27 Where is the project GitHub / live link?

- GitHub — `github.com/Vikash-Kumar87/ALUMNI`
- Frontend (Vercel) — Live URL available from the Vercel dashboard
- Backend (Render) — API URL available from the Render dashboard

28 What are the key environment variables of the project?

Variable	Purpose
`GROQ_API_KEY`	Groq AI API access
`FIREBASE_PROJECT_ID`	Firebase project identifier
`FIREBASE_PRIVATE_KEY`	Firebase Admin SDK auth
`FIREBASE_DATABASE_URL`	Realtime Database URL
`FRONTEND_URL`	CORS allowed origin

29 Summarize the project in one line.

"AlumniConnect is an AI-powered platform that connects students and alumni — career guidance, mentorship, interview practice, and job referrals, all in one place, completely free."

30 What is the industry use case of this project?

- **EdTech startups** — Companies like alumni portals are already raising millions (e.g., AlmaConnect, HiHello)
- **Colleges** — Official alumni management system
- **Corporate HR** — Internal referral and mentorship platforms
- This project's scope directly aligns with products like LinkedIn and Mentorcruise